



REGULATIONS 2024

FLEXIBLE CHOICE BASED CREDIT SYSTEM

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

B.E. Electrical and Electronics Engineering

CURRICULUM

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Exempted for LE	HS	SH1106	19EN106	English for Engineers	-	3	1	4	CGPA	TIP (6 Expts)
2		HS	SH1107	19EN107	Technical Writing and Presentation	-	0	2	2	CGPA	Practical (10 Expts)
3	Exempted for LE	HS	SH9801	NA	Organizational Behavior and Leadership	-	3	0	3	CGPA	Theory
4	Exempted for LE	HS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
5	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு/ Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
6	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
7	Exempted for LE	BS	SH2201	19MA201	Calculus and Matrix Algebra	-	4	0	4	CGPA	Theory
8		BS	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
9	Exempted for LE	BS	SH2223	19MA223	Linear Algebra and Probability	-	4	0	4	CGPA	Theory
10	Exempted for LE	BS	SH3216	19PH216	Physics for Electronics Engineering	-	3	1	4	CGPA	TIP (6 Expts)
11	Exempted for LE	BS	SH4805	NA	Chemistry of Engineering Materials	-	2	1	3	CGPA	TIP (6 Expts)
12	Exempted for LE	BS	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
13		ES	CS3304	19AI304	Fundamentals of C Programming	-	2	1	3	CGPA	TIP (6 Expts)
14	Exempted for LE	ES	CS3305	19AI305	Advanced C Programming	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
15		ES	CS3306	19AI306	Object Oriented Programming using C++	-	2	1	3	CGPA	TIP (6 Expts)
16	Exempted for LE	ES	CS3307	19AI307	Object Oriented Programming using Java	19AI304	2	1	3	CGPA	TIP (6 Expts)
17		ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)
18		ES	CS3408	19AI408	Data Structures	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
19	Exempted for LE	ES	CS3801	NA	Artificial Intelligence and Applications	-	1	2	3	CGPA	Theory (Int Only) and Practical (10 Expts)
20		ES	ME0415	19ME415	Power Generation Engineering	-	4	0	4	CGPA	Theory
21	Exempted for LE	ES	EC2301	19EE301	Electric Circuits and Networks	19MA202	3	1	4	CGPA	TIP (6 Expts)

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
22		PC	EC2801	NA	Electromagnetic Theory	-	3	0	3	CGPA	Theory
23		PC	EC2402	19EE402	Transformers and DC Machines	19EE301,19 EE401	3	1	4	CGPA	TIP (6 Expts)
24		PC	EC2403	19EE403	AC Machines	19EE402	3	1	4	CGPA	TIP (6 Expts)
25		PC	EC2404	19EE404	Digital Electronics	-	3	1	4	CGPA	TIP (6 Expts)
26		PC	EC2405	19EE405	Electronic Circuits	19EE301,19 PH212	3	1	4	CGPA	TIP (6 Expts)
27		PC	EC2802	NA	Measurement and Instrumentation Systems	-	2	1	3	CGPA	TIP (6 Expts)
28		PC	EC2407	19EE407	Linear Integrated Circuits and Applications	19EE405	3	1	4	CGPA	TIP (6 Expts)
29		PC	EC2803	NA	Control Systems	EC2301	2	1	3	CGPA	TIP (6 Expts)
30		PC	EC2804	NA	Electrical Power Transmission and Distribution Systems	EC2301	3	0	3	CGPA	Theory
31		PC	EC2410	19EE410	Power System Analysis	19MA207,19 EE409	3	1	4	CGPA	TIP (6 Expts)
32		PC	EC2411	19EE411	Power Electronics	19PH212	3	1	4	CGPA	TIP (6 Expts)
33		PC	EC2830	NA	Microprocessor and Microcontroller Systems	EC2404	2	1	3	CGPA	TIP (6 Expts)
34		PC	EC2805	NA	Solid State Drives and Control	EC2411	2	1	3	CGPA	TIP (6 Expts)
35		PC	EC2806	NA	Digital Signal Processing	EC2404	3	1	4	CGPA	TIP (6 Expts)
36		PC	EC2415	19EE415	High Voltage Engineering	19EE409	3	0	3	CGPA	Theory
37		PC	EC2807	NA	Fundamentals of Renewable Energy Systems	-	2	1	3	CGPA	TIP (6 Expts)
38		PE(Honours 1)	EC2506	19EE506	Restructured Power Systems	19EE413	3	0	3	CGPA	Theory
39		PE(Honours 1)	EC2504	19EE504	Flexible AC Transmission Systems	19EE413	3	0	3	CGPA	Theory
40		PE(Honours 1)	EC2810	NA	Energy Storage Technologies	-	3	0	3	CGPA	Theory
41		PE(Honours 1)	EC2809	NA	Power Quality	-	3	0	3	CGPA	Theory
42		PE(Honours 1)	EC2812	NA	Microgrid Modelling and Analysis	-	3	0	3	CGPA	Theory
43		PE(Honours 1)	EC2813	NA	Microgrid Protection and Control	-	3	0	3	CGPA	Theory
44		PE(Honours 2)	EC2814	NA	Wind Energy and Sustainability	-	2	1	3	CGPA	TIP (6 Expts)
45		PE(Honours 2)	EC2815	NA	Smart Grid Technology for Renewable Energy System	-	2	1	3	CGPA	TIP (6 Expts)
46		PE(Honours 2)	EC2816	NA	Grid Connected and Standalone PV Solar Plant Design	-	2	1	3	CGPA	TIP (6 Expts)
47		PE(Honours 2)	EC2817	NA	Energy Forecasting and Modelling	-	3	0	3	CGPA	Theory
48		PE(Honours 2)	EC2818	NA	Solar Energy Engineering and Technology	-	3	0	3	CGPA	Theory

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
49		PE(Honours 2)	EC2819	NA	Power Electronics for Renewable Energy Systems	-	2	1	3	CGPA	TIP (6 Expts)
50		PE(Honours 3)	EC2820	NA	Embedded System Design	-	2	1	3	CGPA	TIP (6 Expts)
51		PE(Honours 3)	EC2821	NA	Embedded Control for Electric Drives	-	2	1	3	CGPA	TIP (6 Expts)
52		PE(Honours 3)	EC2822	NA	Embedded System for Automotive Applications	-	2	1	3	CGPA	TIP (6 Expts)
53		PE(Honours 3)	EC2823	NA	Embedded Processors	-	2	1	3	CGPA	TIP (6 Expts)
54		PE(Honours 3)	EC2824	NA	Embedded C- Programming	-	2	1	3	CGPA	TIP (6 Expts)
55		PE(Honours 3)	EC2825	NA	Smart System Automation	-	3	0	3	CGPA	Theory
56		PE(Honours 4)	EC2526	19EE526	Introduction to Fundamentals of Electric Vehicles	-	3	0	3	CGPA	Theory
57		PE(Honours 4)	EC2527	19EE527	Electric Vehicle Powertrain and Integration	19EE526/EC 2526	2	1	3	CGPA	TIP (6 Expts)
58		PE(Honours 4)	EC2528	19EE528	Electric Vehicle Charging Infrastructure	19EE526/EC 2526	3	0	3	CGPA	Theory
59		PE(Honours 4)	EC2529	19EE529	Electric Vehicle Grid Integration	19EE526/EC 2526	2	1	3	CGPA	TIP (6 Expts)
60		PE(Honours 4)	EC2530	19EE530	Embedded IoT for Electric Vehicles	EC2526	2	1	3	CGPA	TIP (6 Expts)
61		PE(Honours 4)	EC2531	19EE531	Embedded AI for Electric Vehicles	19EE526/EC 2526	2	1	3	CGPA	TIP (6 Expts)
62		PE(Honours 4)	EC2532	19EE532	Battery Technologies and Management Systems for Electric Vehicles	19EE526/EC 2526	3	0	3	CGPA	Theory
63		PE(Honours)	EC1410	19EC410	VLSI Design	19EC303 / EC1801 / EC2404	3	1	4	CGPA	TIP (6 Expts)
64		PE(Honours)	EC1523	19EC523	Verilog HDL	19EC303 / EC1801 / EC2404	2	1	3	CGPA	TIP (6 Expts)
65		PE(Honours)	EC1521	19EC521	System Verilog for Design and Verification	19EC523 / EC1523	2	1	3	CGPA	TIP (6 Expts)
66		PE(Honours)	EC1522	19EC522	Physical Design for VLSI	19EC410 / EC1410	2	1	3	CGPA	TIP (6 Expts)
67		PE(Honours)	EC1822	19EC822	VLSI Technology	EC1410 / 19EC410	3	0	3	CGPA	Theory
68		PE(Honours)	EC1517	19EC517	Low Power VLSI Design	19EC410 / EC1410	3	0	3	CGPA	Theory
69		PE(Honours)	EC1519	19EC519	Testing of VLSI Circuits	19EC410 / EC1410	3	0	3	CGPA	Theory
70		PE	CS3404	19AI404	Analysis of Algorithms	19AI301	2	1	3	CGPA	TIP (6 Expts)
71		PE	CS1404	19CS404	Database Management System and its Applications	19AI301/19AI304/19CS301/19CS302/1	3	1	4	CGPA	TIP (6 Expts)
72		PE	CS1509	19CS509	Graph Theory and Applications	19MA206	3	0	3	CGPA	Theory
73		PE	CS3410	19AI410	Introduction to Machine Learning	-	2	1	3	CGPA	TIP (6 Expts)
74		PE	EC2508	19EE508	Power System Transients	19EE415	3	0	3	CGPA	Theory
75		PE	EC2511	19EE511	Protection and Switchgear	19EE403,19EE411,19EE413, EC2804	3	0	3	CGPA	Theory

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
76		PE	EC2512	19EE512	Utilisation, Safety and Maintenance of Electrical Systems	19EE403	3	0	3	CGPA	Theory
77		PE	EC2827	NA	Substation Engineering and Automation	-	3	0	3	CGPA	Theory
78		PE	EC2828	NA	Sustainable Engineering Design for Indoor And Outdoor Illumination	-	2	1	3	CGPA	TIP (6 Expts)
79		PE	EC2509	19EE509	Energy Management and Auditing	-	3	0	3	CGPA	Theory
80		PE	EC2516	19EE516	Computer Aided Design of Electrical Machines	19EE403,19 PH207, EC2403	2	1	3	CGPA	TIP (6 Expts)
81		PE	EC1520	19EC520	Introduction to MEMS and NEMS	-	3	0	3	CGPA	Theory
82		PE	EC2515	19EE515	Soft Computing Techniques	-	3	0	3	CGPA	Theory
83		PE	CS1542	19CS542	Embedded Board Design	-	2	1	3	CGPA	TIP (6 Expts)
84		OE	XXXXXX	19XXXXX	Open Electives	-	0	0	0	CGPA	Theory
85		Online	XXXXXX	19XXXXX	Online Courses	-	0	0	0	CGPA	Experiential Learning
86		EEC	EC2701	19EE701	Mini Project	-	0	2	2	CGPA	Project (Int Only)
87	Option 1	EEC	EC2702	19EE702	Project Work Phase I	-	0	2	2	CGPA	Project (Int Only)
88	Option 1	EEC	EC2703	19EE703	Project Work Phase II	19EE702 / EC2702	0	6	6	CGPA	Project
89	Option 2	EEC	SH6802	NA	Start Up	-	0	8	8	CGPA	Project
90		EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)
91		EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
92		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
93		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 /19EY710	1	0	1	CGPA	Theory (Int Only)
94		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
95		EEC	SH6713	19EY713	Placement Readiness Assessment	19EY704 / 19EY711/SH 6711	1	0	1	CGPA	Theory (Int Only)
96		EEC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)
97		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
98	Any 1	MC	SH8802	NA	Internship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
99	Any 1	MC	SH8803	NA	Entrepreneurship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
100	Any 1	MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
101	Any 1	MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
102	Any 1		SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
------	---------	----------	-------------------	-------------------	--------------	--------------	----------------	-------------------	---------------	-------------------	-------------

BATCH: 2024- 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)
Humanities and Science	11	3
Basic Science	25	6
Engineering Science	26	12
Professional Core	57	57
Professional Electives	15	15
Open Electives	12	12
Employability Enhancement Courses	16	16
Mandatory Courses	3	3
TOTAL	165	124